DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

CLASS TEST-II

SUBJECT:BASIC ELECTRONICS

Class: B.E. (AI&ML)

Semester: III

Academic Year: 2025-26

Max Marks: 30

Duration:1 hr 30 min

Date:26-11-2025

Answer All Questions

PART A (5 x 2 = 10 M)

		Marks	BT	C
1. a)	What are the characteristics of and ideal op – amp?	2	2	
b)	Draw the pin diagram of OP-amp IC	2	2	
c)	What are the advantages of –ve feedback ?	2	2	
d)	What is meant by pinchoff voltage ?	2	1	
e)	Why FET is called as voltage control device justify ?	2	2	

Answer any two

PART B (2 x 10 = 20 M)

2. a)	Derive the general equation for LC oscillator and from that deduce the frequency of oscillation for Hartley Oscillator ?	5	2	
b)	Define g_m,r_d and μ of a JFET and derive the relation between them.	5	3	
3. a)	Explain the working of RC phase shift oscillator and derive the frequency	6	2	
b)	Derive input and output impedence of current series feedback ?	4	4	
4.	Design the circuit for which output will be $V_0=V_1-V_2+2V_3+4V_4$	10	3	